Math 1200 Lecture 4

Sep 15 2026

Topics: 13.3-13.4: Arc length, acceleration and velocity Name:_____
Collaborators:_____

Group work 1 We have seen that the curve parametrized by $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$ traces out a helix. (a) Find the arc length of this helix between the points $(1, 0, 0)$ and $(1, 0, 4\pi)$.

(b) Find the unit tangent vector at the point $(1, 0, 2\pi)$.

Group work 2 Write a formula for the perimeter of an ellipse with axes of length $2a$ and $2b$ in terms of an integral[1].

Group work 3 A particle starts at $(1, 0, 0)$ at time $t = 0$ with initial velocity $\vec{v}(0) = \langle 0, 1, 0 \rangle$ and accelerates in the direction $\vec{a}(t) = \langle t, \cos(t), e^t \rangle$. Where is the particle at $t = 2\pi$?

[1] The resulting integral is famously impossible to solve using elementary functions